Fiscal Year:	FY 2021	**Task Last Updated:**	FY 03/05/2021
PI Name:	Mudawar, Issam Ph.D.		
Project Title:	Flow Boiling and Condensation Experiment (FBCE)		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	West Lafayette	**State:**	IN
Zip Code:	47907-1288	**Congressional District:**	4
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2008 Fluid Physics NNH08ZTT002N
Start Date:	06/01/2017	**End Date:**	12/31/2021
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	3	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
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Flight Program:	ISS		
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Hasan, Mohammad Ph.D. (NASA Glenn Research Center)		
Grant/Contract No.:	NNX17AK98G		
Performance Goal No.:			
Performance Goal Text:			
	EDITOR'S NOTE: Continuation of "Flow Boiling and Condensation Experiment (FBCE)," grants NNX13AB01G and NNX12AK14G, which covered from 6/1/2012-5/31/2017. Several efforts are presently underway to reduce the size and weight of space systems by transitioning from single-phase to two-phase thermal management. Because of large density differences between liquid and vapor, buoyancy can play an important role in defining the motion of vapor relative to liquid and, therefore, influence heat transfer effectiveness. This is particularly the case for future space missions, including the crewed mission to Mars. The emphasis in two-phase fluid physics is evident from several recent NASA workshops that culminated in critical recommendations concerning Rankine cycle power conversion, thermal control systems, and advanced life support systems. Among others, key fluid physics mechanisms that have been identified are flow boiling and condensation. A key strategy is to develop mechanistic models for flow conditions that would ensure insensitivity to gravity level for these important heat transfer		

Task Description:	mechanisms. This would allow performing most of the required confirmatory experiments on Earth. Those topics and the minimum flow requirements constitute key objectives for the proposed study. Unfortunately, most of the two-phase heat transfer know-how amassed over nearly a century of research comes from experiments conducted in Earth's gravity. The proposed study aims to develop an experimentally validated, mechanistic model for the minimum flow criteria to ensure gravity independent flow boiling heat transfer and condensation. Parabolic flight experiments and preparations for International Space Station (ISS) experiments will occur throughout the course of the proposed study. This project will be a joint effort between the Purdue University Boiling and Two-Phase Flow Laboratory (PU-BTPFL) and the NASA Glenn Research Center. Personnel from the two organizations combine extensive experience in research and development of both flow boiling and condensation systems, and in conducting microgravity experiments.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The proposed research will lay the foundation of development of advanced high power to weight ratio thermal management systems for NASA's future space vehicles and planetary bases.
Task Progress:	The present study investigates extensively the capability of computational fluid dynamics (CFD) to predict two-phase fluid flow and heat transfer characteristics of FC-72 flow boiling in vertical upflow. The computational method performs transient analysis to model the entire flow field and bubble behavior for highly subcooled flow boiling in a rectangular channel heated on two opposite walls at high heat flux conditions of about 81% of critical heat flux (CHF). Transient variations of local vapor fraction, thermal energy transfer and concentration, and wall temperature are estimated through the computational methodology based on the Volume of Fluid (VOF) approach combined with shear-lift modeling to overcome a fundamental weakness in modeling multiphase flows. Detailed information about bubble distribution in the vicinity of the heated surface, thermal conduction inside the heating wall, local heat fluxes passing through the solid-fluid interface, and velocity and temperature profiles, which are not easily observed or measured by experiments, is carefully evaluated. Predictions are compared to the experimental data for three sets of operating conditions. Overall, computationally predicted flow pattern, bubble behavior, and heat transfer parameters (such as wall temperature excursion and thermal energy concentration) clearly represent phenomena of premature CHF which take place slightly earlier than actual operating conditions. But, despite these slight differences, the present computational work does demonstrate the ability to effectively predict the severe degradation in heat transfer performance commonly encountered at heat fluxes nearing CHF.
Bibliography Type:	Description: (Last Updated: 01/06/2026)
Articles in Peer-reviewed Journals	O'Neill LE, Mudawar I. "Review of two-phase flow instabilities in macro- and micro-channel systems." International Journal of Heat and Mass Transfer. 2020 Aug;157:119738. https://doi.org/10.1016/j.ijheatmasstransfer.2020.119738 , Aug-2020
Articles in Peer-reviewed Journals	Lee J, O'Neill LE, Mudawar I. "Computational prediction of key heat transfer mechanisms and hydrodynamic characteristics of critical heat flux (CHF) in subcooled vertical upflow boiling." International Journal of Heat and Mass Transfer. 2020 Nov;161:120262. https://doi.org/10.1016/j.ijheatmasstransfer.2020.120262 , Nov-2020
Articles in Peer-reviewed Journals	Zhou L, Garg D, Qiu Y, Kharangate CR, Kim S, Mudawar I. "Machine learning algorithms to predict flow condensation heat transfer coefficient in mini/micro-channel utilizing universal data." International Journal of Heat and Mass Transfer. 2020 Dec;162:120351. https://doi.org/10.1016/j.ijheatmasstransfer.2020.120351 , Dec-2020